

Breaking Barriers:

How AI-Driven Membrane Permeation Science and Exosome Technology Could Redefine Drug Discovery

A Matthew 25 Perspective on Medicines the World Cannot Yet Reach

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EXECUTIVE SUMMARY

- ▶ Scientists found a new way to test medicines for a serious lung disease called tuberculosis. They built a tool called PAC-MAN that checks whether a drug can actually get inside the germ that makes people sick. Then they used AI — a computer that learns from information — to figure out which types of medicines work best.
- ▶ Here is the big idea: a medicine does not just need to be strong. It also has to be able to get through the wall of a germ or a cell to do its job. If it cannot get inside, it does not matter how powerful it is. AI can now help us find medicines that can break through those walls.
- ▶ This article looks at seven kinds of sickness where this same AI tool could help. These include dangerous infections that do not respond to normal medicines, diseases that attack the brain, cancer, and illnesses that mostly hurt poor people around the world who cannot get good health care.
- ▶ Our bodies make tiny natural packages called exosomes. Think of them like little delivery trucks that already know where to go inside your body. Scientists can fill these tiny trucks with the right medicine and send them straight to the sick cells — so the medicine gets exactly where it needs to go.
- ▶ This article proposes a four-step plan that uses AI to pick the best medicine, design the delivery truck, load it correctly, and predict how it will travel through the body. This plan is inspired by Matthew 25 — the call to care for those who are sick and left behind — and is built to help the people who need medicines the most.

The Drug That Never Arrives

In pharmacology, we speak often of efficacy — whether a compound can disable a pathogen, silence a tumor, or slow neurodegeneration. We test binding affinities, enzyme inhibition constants, and minimum inhibitory concentrations with extraordinary precision. But there is a prior question, rarely asked with equal rigor, that determines whether any of that knowledge matters in living tissue: Can the drug actually get there?

A landmark study just published in *Nature Microbiology* by Cox, Bhatt, and colleagues at the Crick Institute offers a new answer — at least for tuberculosis. Using a technique called PAC-MAN (Peptidoglycan Accessibility Click-Mediated Assessment) combined with machine learning cheminformatics, the researchers mapped how 1,572 small molecules traverse the formidable mycomembrane of *Mycobacterium tuberculosis* and built a predictive model — MycoPermeNet — that identifies the molecular structural features most likely to enable permeation and whole-cell antibiotic activity.

Their discovery: nitrogen-containing aromatic scaffolds, particularly indole rings, serve as molecular 'keys' tuned to the specific architecture of the TB outer membrane. The drug does not merely need to be potent. It needs to speak the membrane's language.

"Barrier physics governs therapeutic efficacy more than target affinity alone. A drug must speak the membrane's language before it can do any good inside."

What Makes This Methodology Novel

Before PAC-MAN, measuring drug accumulation inside mycobacterial cells required indirect inferences — observing whole-cell activity and working backwards. The PAC-MAN assay solves this by bioorthogonally trapping molecules at the site of permeation, physically marking where in the membrane architecture a compound accumulates. Combined with click chemistry visualization, it makes the invisible permeation process measurable and quantifiable at scale.

The ML layer — MycoPermeNet — then transforms that data into generalizable structural rules. Given a chemical structure, the model predicts membrane permeation probability and correlates it with whole-cell antibiotic activity. The result is a drug discovery filter that prioritizes not just potency at the target, but the physical capacity to reach it.

This is, at its core, a signal propagation problem solved with machine learning. The biological barrier is a complex, selective filter. The drug molecule is a signal that must carry properties allowing it to traverse that filter. AI identifies which signal properties work — and which do not.

Diseases That Could Be Transformed by the Same Logic

The PAC-MAN/ML framework is not TB-specific. It is a methodology for solving any problem where a therapeutic agent must cross a selective biological barrier to reach its target. That problem is, arguably, one of the most universal in medicine. Below are the highest-impact disease domains where this methodology could be deployed.

1. Drug-Resistant Gram-Negative Bacterial Infections

Pseudomonas aeruginosa, *Acinetobacter baumannii*, and *Klebsiella pneumoniae* — the so-called ESKAPE pathogens — kill hundreds of thousands annually, largely because their outer membranes are nearly impenetrable to conventional antibiotics. The PAC-MAN assay, adapted to Gram-negative lipopolysaccharide membranes, and a species-specific ML model analogous to MycoPermeNet, could identify chemical scaffolds that defeat these barriers. The researchers themselves note that permeation predictors for mycobacteria differ substantially from those for Gram-negative organisms — precisely what makes species-tailored ML models necessary and powerful.

2. *Mycobacterium abscessus* (Pulmonary NTM Disease)

M. abscessus causes progressive, often fatal lung disease — particularly in people with cystic fibrosis — and carries one of the highest intrinsic antibiotic resistance profiles of any bacterial pathogen. Its mycomembrane architecture is nearly identical to *M. tuberculosis*. MycoPermeNet, retrained on *M. abscessus* permeation data, could be deployed with minimal architectural changes and could accelerate drug discovery for a disease that currently has no reliable curative regimen.

3. Invasive Fungal Infections

Candida auris, *Aspergillus fumigatus*, and *Cryptococcus neoformans* represent a growing global crisis, especially in immunocompromised populations. Azole resistance frequently operates through membrane composition changes that reduce drug accumulation inside fungal cells — a problem structurally analogous to mycobacterial membrane impermeability. A PAC-MAN-style assay adapted to fungal cell wall biology, combined with ML structure-permeation modeling, could guide the design of next-generation antifungals with superior intracellular accumulation.

4. Leprosy and Buruli Ulcer

M. leprae and *M. ulcerans* — the causative agents of leprosy and Buruli ulcer — are slow-growing, difficult to culture, and concentrated almost entirely in low-income communities. Because wet-lab drug screens are impractical for these organisms, transfer learning from a MycoPermeNet model trained on *M. tuberculosis* data could allow in silico prioritization of drug candidates without large-scale laboratory infrastructure. This is a Matthew 25 application of the highest order: using AI to do what manual science cannot economically justify.

5. Neurodegenerative Diseases (Blood-Brain Barrier)

The blood-brain barrier (BBB) is the most consequential permeation obstacle in all of medicine. Over 98% of small-molecule drug candidates fail to cross it in therapeutically meaningful concentrations. The indole scaffold identified by MycoPermeNet as a mycomembrane permeator is, notably, already present in serotonin, melatonin, tryptophan, and numerous CNS-active pharmaceuticals — a chemical convergence that hints at shared permeation physics across very different biological membranes.

A PAC-MAN-analogous bioorthogonal assay measuring drug accumulation in brain endothelial cell models, with ML trained on those permeation profiles, could transform drug development for Alzheimer's disease, Parkinson's disease, ALS, and major psychiatric conditions. Parkinson's drug development, in particular — a domain close to my own research and personal experience — could benefit enormously from a permeation-first drug discovery paradigm.

6. Intracellular Cancers

Many oncology drugs fail not from insufficient target affinity but because they cannot penetrate tumor cell membranes or accumulate inside cancer cells at cytotoxic concentrations. Efflux pump overexpression — a resistance mechanism that ejects drugs before they act — is, in chemical terms, the same problem as poor membrane permeation. A PAC-MAN-style accumulation assay applied to cancer cell lines, combined with ML permeation modeling, could identify anti-cancer scaffolds that resist efflux and accumulate intracellularly — reviving compounds that failed for permeation, not potency, reasons.

7. Obligate Intracellular Bacterial Pathogens

Chlamydia trachomatis, *Listeria monocytogenes*, and *Mycobacterium leprae* all hide inside host cells, forcing any drug to cross both the host cell membrane and a bacterial membrane. The layered-barrier physics problem is conceptually identical to the mycomembrane challenge — just compounded. ML models trained to predict sequential permeation across multiple barrier types could be the decisive tool for treating diseases that currently evade therapy by hiding in plain sight inside human cells.

The Exosome Revolution: A New Delivery Architecture

Every disease category above shares a common bottleneck: even if we identify the perfect molecular scaffold for membrane permeation, we still face the challenge of delivering it to the right tissue, the right cell, the right organelle — without destroying healthy tissue along the way. This is where exosome technology enters the picture, not as a complement to the PAC-MAN/ML framework, but as its logical delivery partner.

What Are Exosomes?

Exosomes are nanoscale extracellular vesicles — typically 30 to 150 nanometers in diameter — naturally secreted by virtually all cell types. They are, in essence, the body's own messaging infrastructure: lipid bilayer packets that carry proteins, RNA, DNA fragments, and small molecules between cells, crossing biological barriers that synthetic delivery vehicles cannot. Because they are biological in origin, they are inherently biocompatible, evade immune clearance, and — critically — possess surface proteins that direct them to specific tissues and cell types.

In recent years, researchers have learned to engineer exosomes as drug delivery vehicles: loading them with therapeutic payloads, modifying their surface proteins to target specific cell types, and exploiting their natural barrier-crossing capabilities to reach tissues previously inaccessible to conventional formulations.

Exosomes and the PAC-MAN/ML Framework: A Convergence

Here is where the synergy becomes architecturally profound. The PAC-MAN/ML methodology identifies which molecular scaffolds can permeate a given biological barrier — but it does not specify how to deliver those scaffolds to the site of the barrier in a living organism. Exosomes solve exactly that problem.

Consider the TB case: MycoPermeNet identifies an indole-bearing compound that permeates the mycomembrane with high probability. But *M. tuberculosis* hides inside alveolar macrophages in the lung — cells that are themselves difficult to target with conventional drug formulations. Engineered exosomes derived from macrophages naturally home to lung tissue and fuse preferentially with macrophage membranes. Loading an exosome with a MycoPermeNet-optimized compound creates a two-stage permeation system: the exosome handles Stage 1 (host cell entry), and the ML-optimized scaffold handles Stage 2 (mycobacterial membrane permeation).

"Exosomes are the body's own delivery network. Combining them with AI-optimized permeation scaffolds creates a two-stage system — biological intelligence paired with machine intelligence — to reach pathogens that hide where conventional medicine cannot follow."

Disease-Specific Exosome Applications

- ▶ **Tuberculosis & *M. abscessus*** — Macrophage-derived exosomes engineered to carry MycoPermeNet-optimized indole scaffolds. The exosome delivers the compound directly into infected macrophages; the scaffold then permeates the mycobacterial membrane with AI-predicted efficiency.
- ▶ **Neurodegeneration (Alzheimer's, Parkinson's)** — Exosomes naturally cross the blood-brain barrier — a feat that 98% of conventional drugs cannot achieve. Brain-targeting exosomes loaded with BBB-permeation-optimized small molecules (identified by a PAC-MAN-analogous ML model for CNS drug discovery) could deliver therapeutic concentrations to neurons that are currently unreachable.
- ▶ **Drug-Resistant Gram-Negative Infections** — Neutrophil-derived exosomes, which naturally accumulate at sites of bacterial infection, could carry Gram-negative membrane-permeation-optimized compounds (identified by a Gram-negative-specific ML model) directly to sites of *P. aeruginosa* or *A. baumannii* infection — including biofilm-embedded organisms that resist conventional antibiotic penetration.
- ▶ **Intracellular Cancer** — Tumor-derived exosomes exhibit a remarkable property: they preferentially re-target to their cell of origin, a phenomenon called homotypic targeting. Exosomes derived from cancer cell lines can carry ML-optimized anti-cancer scaffolds back to tumor cells with extraordinary selectivity, reducing off-target toxicity.
- ▶ **Fungal Infections** — Exosomes derived from macrophages or dendritic cells — the immune cells that first encounter fungal pathogens — could carry antifungal compounds optimized for fungal membrane permeation directly to sites of infection, including the deep-tissue disseminated infections that are currently most difficult to treat.

The AI Layer: Engineering Smarter Exosomes

Exosome engineering is itself becoming an AI-assisted process. Machine learning models are now being trained on exosome surface proteomics data to predict which surface modifications maximize targeting specificity for a given tissue or cell type. Combining this with MycoPermeNet-class permeation modeling creates an end-to-end AI-guided drug delivery pipeline:

- ▶ **Step 1 — Payload Optimization** — ML models identify molecular scaffolds with optimal permeation properties for the target barrier (mycobacterial membrane, BBB, fungal cell wall, tumor cell membrane).
- ▶ **Step 2 — Exosome Engineering** — AI models identify surface protein modifications that maximize exosome targeting to the tissue/cell type harboring the pathogen or cancer.
- ▶ **Step 3 — Loading Optimization** — ML models trained on exosome cargo loading data identify the formulation conditions that maximize payload encapsulation efficiency and stability.
- ▶ **Step 4 — Delivery Prediction** — Pharmacokinetic AI models simulate the in vivo journey from administration to target cell, predicting tissue distribution and optimizing dosing.

This is not speculative architecture. Each of these steps is being pursued in laboratories today. What is still missing is the integration layer — the AI framework that connects payload optimization, exosome engineering, loading, and delivery prediction into a unified drug design pipeline. That integration is a design challenge directly addressable by the QAIV SiP framework.

A Signal Propagation View: The QAIV Connection

From the perspective of the AI Vibrations Theory and the QAIV (Quantum AI Vibrations) framework, the PAC-MAN/ML + exosome architecture is a deeply familiar structure: a multi-layer signal propagation system in which a message (the therapeutic molecule) must traverse a series of selective filters (biological membranes, immune clearance mechanisms, tissue barriers) before reaching its target receiver (the pathogen or diseased cell).

MycPermeNet does what any good signal optimization model does: it identifies the carrier properties — in this case, molecular scaffold features like the indole ring — that maximize signal transmission across a specific filter. The exosome is the outer envelope of that signal: a biological carrier that handles the macro-level routing problem, navigating the body's own communication infrastructure to position the signal near its target before releasing it.

This is the same architecture that governs wireless communications, neural signaling, and quantum information transmission. The mathematics of barrier permeation, carrier optimization, and targeted delivery are structurally isomorphic across these domains. AI that can learn the rules of one can, in principle, be adapted to learn the rules of others — which is precisely what MycPermeNet demonstrates for biology.

"The indole scaffold functions like a resonant frequency — a molecular signature tuned to pass through the mycomembrane's selective filter. This is physics, biology, and AI converging on the same truth: communication requires a carrier whose properties match the barrier it must cross."

A Matthew 25 Imperative

Tuberculosis kills 1.3 million people annually. Nearly all of them are poor. Leprosy, Buruli ulcer, M. abscessus — the mycobacterial diseases most amenable to a PAC-MAN/ML pipeline — are concentrated almost entirely in communities that have never had the economic or political weight to drive pharmaceutical R&D investment. The BBB problem disproportionately affects diseases of aging in populations that are rapidly expanding in the Global South.

The PAC-MAN/ML + exosome convergence is not merely a scientific opportunity. It is a moral one. When AI can transform drug discovery from a billion-dollar, decade-long empirical process into a model-guided,

targeted design workflow — and when exosomes offer a delivery architecture that reduces off-target toxicity and improves therapeutic efficiency — the barriers to bringing effective medicines to neglected populations begin, finally, to fall.

Matthew 25:36 — 'I was sick and you looked after me.' In the language of AI solutions architecture, this looks like: a PAC-MAN assay generating permeation training data, a MycoPermeNet-class model identifying the molecular scaffolds that work, an exosome engineering pipeline delivering them to the tissues that harbor disease, and an AI integration layer connecting every step — designed explicitly in service of the people the pharmaceutical market has historically left behind.

That is the work. That is the vision. That is what it means to build AI for the common good.

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